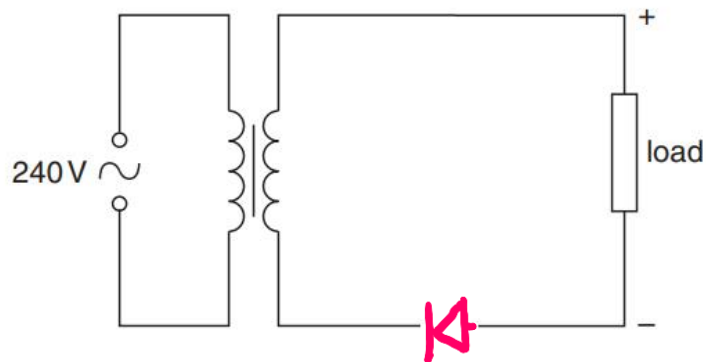


6 (a)(i) Iron core prevent flux losses or improve flux linkage B1

(a)(ii) Primary a.c. current causes changing flux in iron core  
e.m.f. / current (induced) in core B1

induced/eddy current in core causes (joule)heating B1  
(b)(i) diode correctly drawn to give polarity across load. B1



(b)(ii)  $\frac{N_s}{N_p} = \frac{V_s}{V_p}$  C1  
C1

$$V_0 = \sqrt{2} \times V_{\text{rms}} = \sqrt{2} \times 240$$

$$\text{ratio} = 9.0 / (\sqrt{2} \times 240)$$

$$= 1/38 \text{ or } 0.027$$

A1

(b)(iii)  $V_{\text{rms}} = 9.0/2 = 4.5 \text{ V}$  C1

$$\text{Mean power output} = V_{\text{rms}}^2/R = (4.5)^2/4.5$$
$$= 4.5 \text{ W}$$

A1

